

Use Case Descriptions: UC06 to UC10

UC06: Lab Result Integration and Notification

Field	Detail
Use Case ID	UC06
Use Case Name	Lab Result Integration and Notification
Actors	Lab Staff (Primary), System, Ward Doctor, OPD Doctor
Related FRs	FR06, FR07

Preconditions

- A lab test request exists in the system (UC05).
- Lab staff are authenticated in the lab portal.
- The specimen has been collected and analysis is complete.

Main Flow

1. Lab Staff opens the pending requests worklist in the lab portal.
2. System displays requests sorted by urgency (STAT → Urgent → Routine) and by age (oldest first within each tier).
3. Lab Staff selects a request and enters the completed test results using the data model derived from the observed lab slip (DC06):
Parameter name (e.g., WBC, Lymphocytes, Neutrophils)
Result value (numerical)
Unit (e.g., $10^3/\mu\text{L}$)
Reference Range: low value and high value (e.g., 4.00 10.00)
WBC Message / interpretive flag (free text)
4. System validates result format and automatically checks all values against critical threshold ranges.
5. Lab Staff reviews and confirms the result entry.
6. System marks the request status as "Complete" and appends the result to the patient's longitudinal medical record (UC09).
7. System immediately dispatches a notification to the requesting doctor (UC07): "Results ready [Patient Name] [Test Name] [Request ID]."
8. Result becomes visible in the doctor's patient record view with a "New Result" badge.

Alternative Flows

AF06A: Critical Value Detected

- At step 4, if a result value falls outside a predefined critical threshold (e.g., $\text{WBC} < 2.0$ or $> 30.0 \times 10^3/\mu\text{L}$):
 - System flags the result as "Critical" before the lab staff can confirm submission.

- Lab Staff must explicitly acknowledge the critical flag.
- System dispatches an escalated P1 notification to the requesting doctor: "CRITICAL RESULT Immediate Review Required [Patient Name]."
- System logs: time result was entered, time doctor acknowledges, and time doctor accesses the record.

AF06B: Result Requires Amendment

- After step 6, if a lab staff member identifies an error in an already submitted result:
 - Lab Staff selects "Amend Result," enters corrected values, and provides a mandatory amendment reason.
 - System preserves the original result in the audit trail and appends the amendment as a new version.
 - System renotifies the requesting doctor: "Amended result available [Patient Name] [Test Name]."

AF06C: Specimen Rejected

- At step 3, if the specimen is unsuitable (haemolysed, insufficient volume, wrong container type):
 - Lab Staff marks the request "Specimen Rejected" and selects a rejection reason from a dropdown.
 - System notifies the requesting doctor and the ward nurse to arrange recollection.
 - The original request remains open pending a new specimen.

Postconditions

- Completed lab result is stored in the patient's longitudinal medical record.
- Requesting doctor has been notified in real time.
- Full audit trail captures: who entered the result, when, whether amendments were made, and when the doctor acknowledged.

UC07: RealTime Notification System

Field	Detail
Use Case ID	UC07
Use Case Name	RealTime Notification System
Actors	System (Primary), All Clinical Staff (Recipients)
Related FRs	FR07

Preconditions

- A triggering system event has occurred (lab result ready, medication overdue, patient admitted, critical value detected, discharge requested).
- The intended recipient is a registered system user with an active session or a push notification subscription.

Main Flow

9. A system event triggers a notification (sourced from UC01, UC04, UC05, UC06, UC08, or UC10).
10. System identifies the notification type and assigns a priority level:
 - P1 Critical: Critical lab values; emergency admissions; medication missed > 30 minutes.
 - P2 Urgent: STAT lab results ready; patient discharge pending doctor approval.
 - P3 Standard: Routine results ready; new medication orders placed; admission confirmations.
11. System identifies the intended recipient(s) based on role, ward assignment, or explicit doctorpatient link.
12. System dispatches the notification via inapp alert; P1 and P2 also trigger a push notification to the recipient's registered device.
13. Notification displays: patient name, event type, timestamp, and a deep link directly to the relevant record.
14. Recipient acknowledges the notification by tapping or clicking.
15. System records the acknowledgement timestamp and clears the notification from the active queue.

Alternative Flows

AF07A: Recipient Not Logged In (P1 Critical)

- At step 4, if the primary recipient is not active in the system:
 - System escalates to the next available clinician of the same role in the same ward.
 - System logs the failed delivery attempt and the escalation action.
 - If no active recipient is found within 5 minutes, system alerts the Ward Manager.

AF07B: Recipient Acknowledges but Does Not Act (P1 Critical)

- At step 6, if a P1 notification is acknowledged but the patient record is not accessed within 10 minutes:

- System sends a followup alert: "Unreviewed critical result [Patient Name] [Test]."
- System escalates to the Ward Manager after a further 5 minutes of inaction.

AF07C: Quiet Hours P3 Suppression

- During configurable quiet hours (e.g., 23:0006:00):
 - P3 Standard notifications are batched and delivered as a single summary at the end of the quiet period.
 - P1 Critical and P2 Urgent notifications are never suppressed, regardless of quiet hour settings.

Postconditions

- All notifications are logged: trigger event, intended recipient, dispatch time, acknowledgement time.
- Unacknowledged P1/P2 notifications trigger an escalation chain.
- The full notification audit trail is available for clinical governance review.

UC08: Patient Discharge

Field	Detail
Use Case ID	UC08
Use Case Name	Patient Discharge
Actors	Ward Doctor (Primary), Nurse, System
Related FRs	FR08, FR02, FR09

Preconditions

- Patient is currently admitted with status "Occupied [Bed]."
- Ward Doctor has clinically assessed the patient as ready for discharge.
- All outstanding lab results have been reviewed or explicitly acknowledged (UC06).
- No active unresolved medication orders remain (UC04).

Main Flow

16. Ward Doctor opens the patient's record and selects "Initiate Discharge."
17. System checks for open items: pending lab results, unacknowledged critical alerts, active medication orders.
18. If any exist, system displays a checklist before proceeding.
19. Ward Doctor resolves or explicitly acknowledges all open items.
20. System presents the discharge summary form. Ward Doctor completes:
 - Final Diagnosis with ICD10/11 code (DC07)
 - Comorbidities with ICD codes (DC07)
 - Procedure / Intervention / Surgery with ICD code (DC07)
 - Treatment summary and takehome medications
 - Followup instructions
21. Ward Doctor selects the patient outcome from Health 26 Section D categories (DC03):
 - Live Discharge: Referred to Clinic / Referred to Higher Level / Back Referral / No Followup Required
 - Death: Within 24 hours / After 24 hours
 - Missing From Ward
 - Self Discharged
 - Transfer Out
22. Ward Doctor completes the Disease Notification field (DC08): Notifiable Disease Yes/No; if Yes, Date of Notification.
23. Ward Doctor submits the discharge summary.
24. System notifies the duty nurse to prepare the patient for departure (UC07 P2).
25. Nurse confirms physical discharge: patient has left the ward.
26. System updates patient status to "Discharged" and records the discharge timestamp.
27. System releases the bed back to the available pool immediately (UC02 realtime update).
28. Discharge summary is appended to the patient's permanent medical history (UC09).

Alternative Flows

AF08A: Open Items Block Discharge

- At step 2, if one or more open items are detected:
 - System displays the outstanding checklist (e.g., "2 lab results pending review").
 - Ward Doctor must resolve or explicitly acknowledge each item with a reason before the discharge form is unlocked.

AF08B: Discharge Against Medical Advice (DAMA)

- At any point after step 1, if the patient insists on leaving against medical advice:
 - Ward Doctor selects "Discharge Against Medical Advice."
 - System requires mandatory documentation: reason, statement that patient was informed of risks, and witness name.
 - Discharge is recorded with a DAMA flag, visible in all future admissions.

AF08C: Transfer Instead of Discharge

- At step 5, if "Transfer Out" is selected:
 - System prompts for: receiving facility name, transfer reason, and transport mode.
 - Discharge summary is generated as a transfer summary with the receiving institution identified.
 - Bed is released at the point of physical departure confirmation.

Postconditions

- Patient record is closed for the current admission and status is set to "Discharged."
- Bed is returned to the available pool immediately.
- Discharge summary (including ICD codes, outcome, disease notification, and medications) is stored in the patient's permanent medical history.

UC09: Access Complete Patient Medical History

Field	Detail
Use Case ID	UC09
Use Case Name	Access Complete Patient Medical History
Actors	Ward Doctor (Primary), OPD Doctor, Nurse, Lab Staff (ReadOnly)
Related FRs	FR09

Preconditions

- The patient is registered in the system (from any prior interaction).
- The requesting clinician is authenticated with a role that permits record access.
- If the patient is new, a record will be created at first interaction.

Main Flow

29. Clinician searches for the patient by: full name, persistent patient ID, NIC number, or BHT Number (DC01: BHT search supported for backward compatibility with paper records).
30. System returns matching records; clinician selects the correct patient.
31. System displays the patient's complete medical history in a structured chronological timeline:
 - All past admissions and discharge summaries, each referenced by BHT Number
 - All diagnoses (provisional and confirmed) with ICD codes
 - All medication orders and administration records (with nurse IDs and timestamps per DC05)
 - All lab test requests and results (Parameter, Result, Unit, Reference Range per DC06)
 - Allergy and adverse drug reaction records (Drug / Food / Other from Section E, per DC04)
 - Doctor observations and treatment updates, including all Section C vitals
32. Clinician applies filters (date range, event type, department) to navigate the timeline.
33. Clinician views the relevant sections to support clinical decisionmaking.
34. System logs the access event: user ID, role, timestamp, patient record accessed, sections viewed.

Alternative Flows

AF09A: New Patient No Prior Record

- At step 2, if no records match the search:
 - System prompts: "No record found. Register new patient?"
 - Clinician confirms; system creates a new patient record capturing demographics.
 - The new record begins with the current encounter.

AF09B: Restricted Record Access

- At step 3, if the patient record carries a confidentiality restriction (e.g., sensitive diagnosis category):
 - System shows summarylevel information only to standard roles.
 - Full record access requires the clinician to confirm a "Clinical Need to Access" justification.
 - The justification is logged and flagged for periodic audit.

AF09C: Record Locked by Active Edit

- At step 3, if another clinician is actively editing the record:
 - System displays the record in readonly mode with a banner: "Currently being edited by [Doctor Name]."
 - Clinician may request a notification when the record becomes available for editing.

Postconditions

- All record access events are logged in an immutable audit trail.
- The clinician has the full clinical context needed for safe decisionmaking.
- No record content is altered unless the clinician performed an explicit edit action.

UC10: Ward Round Recovery Status Update

Field	Detail
Use Case ID	UC10
Use Case Name	Ward Round Recovery Status Update
Actors	Ward Doctor (Primary), Nurse, System
Related FRs	FR03, FR10

Preconditions

- Patient is admitted and has an active inpatient record.
- Ward Doctor is conducting a scheduled or unscheduled ward round.
- Ward Doctor is authenticated and the patient appears on their ward round list.

Main Flow

35. Ward Doctor opens the ward round list a consolidated view of all assigned patients sorted by priority score and lastreview timestamp.
36. Ward Doctor selects a patient to review.
37. System displays the patient's current summary: last observation, active medications, outstanding lab results, and any nurse notes entered since the last round (AF10C).
38. Ward Doctor conducts the bedside assessment and enters the updated recovery status:
Recovery trajectory: Improving / Stable / Deteriorating / Critical
Updated vitals and clinical notes (using the same Section C fields as UC03)
Any changes to the treatment plan (triggers UC03)
39. Ward Doctor sets the next review interval (e.g., "Review in 4 hours," "Review tomorrow morning").
40. System saves the update, timestamps it, and appends it to the patient's longitudinal record.
41. System notifies the duty nurse of any new instructions arising from the ward round (UC07 P3).
42. System updates the ward round list: lastreviewed time refreshes and priority score recalculates based on recovery trajectory and time since last review.

Alternative Flows

AF10A: Patient Status Deteriorating Escalation Required

- At step 4, if the Ward Doctor selects "Deteriorating" or "Critical":
 - System immediately dispatches a P1 notification to the oncall senior doctor (UC07).
 - System adds a red visual flag to the patient's entry on the ward round list.
 - Ward Doctor is prompted to document an escalation plan before saving.

AF10B: Ward Round Interrupted

- At any step, if the Ward Doctor is called away before completing the review:
 - System autosaves any partial notes as a draft, clearly labelled "Draft Round Incomplete."

- The patient remains at the top of the ward round list with a "Review In Progress" indicator.
- On return, the Ward Doctor resumes from the saved draft.

AF10C: Nurse Adds PreRound Observation

- Before step 1, a nurse may add a preround observation note to a patient's record.
 - The note is surfaced prominently at step 3 as "Nursing Note PreRound [Nurse Name] [Time]."
 - Nurse notes are readonly for the Ward Doctor; they supplement but do not replace doctor observations.

Postconditions

- Recovery status is updated in the patient's longitudinal record with an immutable timestamp.
- Ward round list reflects the updated review time and recalculated priority score.
- Nursing staff are notified of any changes arising from the round.
- Recovery trajectory across multiple rounds is available for trend analysis and clinical governance.